ARDUINO CODE

```
#include <SoftwareSerial.h>
#define TRIG 2
#define ECHO 3
#define ENA 5
#define ENB 6
#define IN1 8
#define IN2 9
#define IN3 10
#define IN4 11

SoftwareSerial BT(12, 13); // RX, TX (Bluetooth)

long duration;
int distance;

void setup() {
  pinMode(TRIG, OUTPUT);
  pinMode(ECHO, INPUT);
  pinMode(ENA, OUTPUT);
  pinMode(ENB, OUTPUT);
  pinMode(IN1, OUTPUT);
  pinMode(IN2, OUTPUT);
  pinMode(IN3, OUTPUT);
  pinMode(IN4, OUTPUT);
  BT.begin(9600);
  Serial.begin(9600);
}

void loop() {
  digitalWrite(TRIG, LOW); delayMicroseconds(2);
  digitalWrite(TRIG, HIGH); delayMicroseconds(10);
  digitalWrite(TRIG, LOW);
  duration = pulseIn(ECHO, HIGH);
  distance = duration * 0.034 / 2;

  if (BT.available()) {
    char cmd = BT.read();
    if (cmd == 'F') forward();
    else if (cmd == 'B') backward();
    else if (cmd == 'L') left();
    else if (cmd == 'R') right();
    else if (cmd == 'S') stopMotors();
  }

  if (distance < 20) stopMotors(); // Safety stop on obstacle
}
```

```
void forward() {
  digitalWrite(IN1, HIGH); digitalWrite(IN2, LOW);
  digitalWrite(IN3, HIGH); digitalWrite(IN4, LOW);
  analogWrite(ENA, 200); analogWrite(ENB, 200);
}

void backward() {
  digitalWrite(IN1, LOW); digitalWrite(IN2, HIGH);
  digitalWrite(IN3, LOW); digitalWrite(IN4, HIGH);
}

void left() {
  digitalWrite(IN1, LOW); digitalWrite(IN2, HIGH);
  digitalWrite(IN3, HIGH); digitalWrite(IN4, LOW);
}

void right() {
  digitalWrite(IN1, HIGH); digitalWrite(IN2, LOW);
  digitalWrite(IN3, LOW); digitalWrite(IN4, HIGH);
}

void stopMotors() {
  digitalWrite(IN1, LOW); digitalWrite(IN2, LOW);
  digitalWrite(IN3, LOW); digitalWrite(IN4, LOW);
}
```

CIRCUIT DIAGRAM

FlexiClean — Circuit Diagram (diagrammatic)

